Step 2 : Data

The expected frequency for each digit = $\frac{10000}{10} = 1000$

Step 3 : Level of significance $\alpha = 5\%$

Step 3 : Test statistic

The test statistic is computed using the following table.

Observed frequency (O_i)	Expected frequency (E_i)	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
1026	1000	26	676	0.676
1107	1000	107	11449	11.449
997	1000	3	9	0.009
966	1000	34	1156	1.156
1075	1000	75	5625	5.625
933	1000	67	4489	4.489
1107	1000	107	11449	11.449
972	1000	28	784	0.784
964	1000	36	1296	1.296
853	1000	147	21609	21.609
			Total	58.542

The test statistic is calculated as

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$= 58.542$$

Step 4 : Critical value

Critical value for 9 df at 5% level of significance is 16.919 i.e., $\chi^2_{9,0.05} = 16.919$

Step 5 : Decision

Since the calculated χ_0^2 (58.542) is greater than the critical value $\chi^2_{9,0.05} = 16.919$, reject H_0. Hence, the digits are not uniformly distributed in the directory.

POINTS TO REMEMBER

- If the number of elements in the sample is less than 30, it is called a small sample.
- For conducting t-tests the parent population(s) should be normal and the samples(s) should be small.
- In case of two sample problems based on t-distribution the sizes of both samples must be less than 30.
- The t-distribution is symmetrical about its mean(zero)
- When the degrees of freedom is large the t-distribution converges to $N(0,1)$ distribution.

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